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PATENTABILITY DISCLOSURE

TankGuard: An IoT-Enabled In-Tank Water Quality Monitoring and Mains-Isolated Pump-Control System for Domestic and Institutional Storage Tanks

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1. TITLE OF INVENTION

"A self-powered IoT-based in-tank water quality monitoring system with edge-computed composite scoring, trend prediction and a local-network dashboard, paired with a mains-isolated home pump controller for domestic and institutional water storage tanks"

2. FIELD OF THE INVENTION

The present invention relates to the field of Internet of Things (IoT) enabled water storage tank maintenance and water quality management. More particularly, the invention pertains to a self-powered IoT sensing and edge-computing unit that is removably installed within a domestic or institutional water storage tank (overhead or underground) and continuously measures key water quality parameters of the stored water, together with a separate indoor pump-control unit. The tank unit computes at the network edge a composite water quality score indicative of accumulated contamination, predicts degradation from historical trends, and raises a prioritised cleaning alert; the home unit displays that status and switches the pump through an opto-isolated relay and a motor-rated contactor, with the electronics never touching mains.

> **IoT Characterisation of the Invention:** The invention is an edge-computing IoT system in which (i) a battery of physical sensors forms the perception layer, (ii) an on-device microcontroller forms the edge/fog computing layer where scoring, trend analysis and event classification are executed locally, (iii) a Wi-Fi access point + embedded web server forms the local communication layer serving an HTML dashboard to any browser on the LAN, and (iv) a separate, mains-isolated home unit forms the pump-control layer, using an opto-isolated relay and a motor-rated contactor with fail-safe OFF. **No cloud, no subscription and no internet uplink is required for any core function** — a deliberate architectural distinction over prior-art IoT water monitors.

3. BACKGROUND AND PROBLEM STATEMENT

3.1 The Forgotten Tank Problem

Water storage tanks in residential households, schools, hospitals, hostels, apartment complexes, and commercial buildings across India and similar tropical regions are among the most neglected components of the safe drinking water chain. Municipal or borewell water entering these tanks is typically treated and relatively clean at the point of entry.

However, water that is stored in tanks for hours to weeks undergoes progressive degradation through the following mechanisms:

1. **Dissipation of disinfectant residual** — Free chlorine, which inhibits microbial growth, dissipates within 24 to 48 hours of storage, removing the primary safeguard against bacterial proliferation.
2. **Sedimentation of suspended particles** — Fine particulate matter (silt, clay, corrosion products, organic detritus) carried in by the source water settles under gravity onto the tank floor, forming a persistent sludge layer that is not removed by normal outlet flow.
3. **Biofilm formation** — Microbial colonies colonize the submerged internal surfaces (walls, floor, baffles, inlet pipe exteriors), forming a slimy biofilm matrix that periodically sheds cells and fragments into the stored water.
4. **Algal growth** — In transparent or semi-transparent tanks exposed to sunlight, algae proliferate on illuminated surfaces and in the water column, particularly during warm months.
5. **Chemical changes** — pH may drift due to CO_2 absorption, pipe corrosion, or biological activity; dissolved solids may leach from accumulated debris.

3.2 The Detection Gap

Currently, tank owners detect water quality degradation through three unsatisfactory methods:

1. **Olfactory/gustatory detection** — By the time the water smells "off" or tastes foul, significant microbial proliferation and/or biofilm shedding has already occurred. The consumer has already been exposed to degraded water.
2. **Visual detection** — Discoloration or visible cloudiness indicates advanced contamination (turbidity typically exceeding 5 NTU), long after the degradation process began.
3. **Calendar-based cleaning** — Tanks are cleaned on arbitrary schedules (monthly, quarterly, annually) regardless of actual contamination state, leading to either insufficient cleaning (health risk) or excessive cleaning (wasted water, labor, cost).

3.3 The Root Cause: Settled Sediment That No One Addresses

The fundamental physical reality that existing solutions fail to address is that the **primary contamination reservoir in a stored-water tank is the settled sludge layer at the tank floor**. This layer:

- Accumulates continuously from incoming water particulates
- Provides a nutrient-rich substrate for anaerobic bacterial growth
- Re-suspends turbulently during inlet refilling, temporarily spiking water column turbidity
- Is not removed by outlet draw-off (outlet is typically above the sludge layer)
- Requires manual drain-and-scrub cleaning, which is labor-intensive, water-wasting, and infrequently performed

3.4 Limitations of Existing Approaches

Existing patents and products in this domain suffer from one or more of the following limitations:

- **Monitoring without prioritisation:** Sensor-only systems (e.g., WO2024173389A1) detect water quality changes but reduce them to raw readings or simple threshold alerts; the owner is given no composite, trend-aware judgement of when cleaning is actually due.

- **Full-tank drain cleaning:** Automated cleaning systems (e.g., IN202441068267) use mechanical wipers and drain the entire tank, wasting large volumes of water and requiring the tank to be out of service.
- **Industrial-scale solutions:** Sludge discharge systems (e.g., CN223988161U) are designed for large municipal sedimentation tanks using camera-based image analysis, not for domestic or institutional storage tanks.
- **In-pipe / point-of-use monitoring:** Systems like WO2024173389A1 and Delta Faucet's application monitor water in pipes or at faucets, not in the storage tank where degradation actually occurs.
- **Chemical treatment approaches:** Systems that dispense chemicals (e.g., chlorine) address microbial contamination but do not remove the physical sludge reservoir that sustains microbial growth.
- **Cloud dependence:** Most IoT-based approaches require internet connectivity and cloud services, adding subscription cost and privacy concerns.

4. SUMMARY OF THE INVENTION

4.1 Core Inventive Concept

The present invention provides a **two-unit system**: a self-powered tank unit that combines continuous in-tank water quality sensing with composite contamination scoring and trend prediction, and a separate home unit that displays that status and controls the pump with full mains isolation, both operating entirely locally without cloud dependency.

The key inventive step is the **edge-computed, trend-predictive health assessment coupled to a fail-safe pump controller**:

1. A multi-parameter sensor array submerged in the stored water continuously measures turbidity, pH, total dissolved solids (TDS), temperature, and water level.
2. An onboard processor computes a **composite TankScore** — a single weighted, trend-aware index that quantifies the overall contamination state of the stored water, with particular emphasis on indicators of settled-sediment activity (turbidity trend, TDS leaching rate, refill-induced resuspension events).
3. When the TankScore crosses a predetermined "clean-required" threshold (or when a manual confirmation is received from the user interface), the system raises a persistent, prioritised cleaning alert and logs the event, so that manual drain-and-scrub cleaning is scheduled on evidence rather than on a calendar.
4. The same tank unit wirelessly publishes level and health status to a separate indoor home unit. That home unit drives the pump through an opto-isolated relay whose contacts energise a motor-rated AC-3 contactor — the low-voltage electronics never make contact with mains.
5. The pump controller enforces fail-safe behaviour: the relay is de-energised on processor reset, and the pump is forced OFF if the tank link is lost for more than a configured timeout. A single local button provides manual override with automatic release.
6. A complete event log (samples, TankScore history, cleaning confirmations, pump run events) is stored locally and presented on a user-accessible dashboard, enabling the owner to track long-term contamination trends and pump behaviour.

4.2 Novel Aspects (Distinguishing from Prior Art)

Novel Aspect	Distinguishing Feature

1. Trend-predictive TankScore	The trigger is not a simple threshold crossing but a weighted composite score incorporating rate-of-change (14-day slope), acceleration, and refill-response analysis of turbidity and TDS. This enables actionable alerts BEFORE water quality becomes visibly degraded.
2. Integrated self-powered sensing unit	All sensing, edge processing, power management and the local dashboard are contained in a single tank-lid-mounted unit with onboard solar panel and battery. No external wiring, no tank penetration, no cloud dependency.
3. Sludge-resuspension event detection	The system distinguishes between (a) gradual degradation (biofilm growth), (b) sudden contamination from source water (refill events), and (c) sludge resuspension (inlet turbulence). Each has a different TankScore contribution and different remediation strategy. This multi-event classification is novel.
4. Local-only, subscription-free architecture	Dashboard, history, and alerts are served from the device itself over local WiFi (mDNS). No cloud service, no account, no subscription. This is a deliberate architectural choice that distinguishes from all IoT-based prior art which requires cloud platforms.

5. DETAILED DESCRIPTION OF EMBODIMENTS

5.1 System Architecture Overview

The invention is embodied as a tank-lid-mounted enclosure containing the sensing electronics, with sensor probes connected externally by waterproof cables, plus a separate indoor pump-control enclosure. The architecture follows a layered IoT stack:

LAYER 4 — PUMP CONTROL (separate indoor unit)
ESP8266 + TFT display + local button → opto-isolated relay → motor-rated AC-3 contactor → pump. Fail-safe OFF on reset or link loss; no mains contact by the electronics.

LAYER 3 — EDGE COMPUTING (on-device, no cloud)
ESP32 dual-core MCU: sensor fusion → TankScore engine → trend regression → event classification → alert decision logic → embedded web server (HTTP + mDNS) → event logging to flash

LAYER 2 — LOCAL COMMUNICATION (LAN only)
ESP32 Wi-Fi access point / station mode → serves HTML5 dashboard to any browser on the same LAN → local notifications, no internet uplink required

LAYER 1 — PERCEPTION (sensors in water)
Turbidity (SEN0189) · TDS/EC (XKC-Y25) · pH (glass electrode + ADS1115 16-bit ADC) · Temperature (DS18B20, 1-Wire) · Level (JSN-SR04T ultrasonic)

LAYER 0 — POWER (self-sustaining)
5–10 W solar panel → charge controller → Li-ion/$LiFePO_4$ battery → regulated rails: 12 V battery bus / 5 V (sensors) / 3.3 V (MCU)

5.2 Component List (Bill of Materials)

The invention is implemented using commercially available off-the-shelf components. The following table identifies the components of a working embodiment, their function, and representative part numbers:

Component	Representative Part	Interface	Function in the Invention
Microcontroller (edge processor)	ESP32-WROOM-32 DevKit	Wi-Fi + BT, GPIO, I²C, 1-Wire, ADC	Runs all firmware: measurement scheduling, TankScore computation, trend regression, event classification, embedded HTTP server, mDNS discovery, flash logging, deep-sleep power management
External ADC	ADS1115 16-bit, 4-channel (I²C)	I²C @ 0x48	Digitises the high-impedance pH electrode output (0–2 V differential) with 16-bit resolution — the ESP32's internal 12-bit ADC is too coarse/noisy for mV-level pH measurements
Turbidity sensor	DFRobot SEN0189 (analog, IP68)	Analog → ADS1115 or ESP32 ADC	90° nephelometric infrared scattering; primary indicator of suspended particles, biofilm fragments and sludge resuspension; 40% of TankScore

TDS sensor	XKC-Y25-NPN (analog + temperature)	Analog (0.2–2.0 V)	Two-electrode conductivity with NTC compensation; indicates dissolved-solids leaching from settled debris; 25% of TankScore
pH sensor	PH-01 glass electrode + op-amp front end	Analog via ADS1115; 5 V excitation + 10k/10k divider to 3.3 V	Potentiometric pH measurement; indicates algal activity, anaerobic decay and corrosion; 20% of TankScore
Temperature sensor	DS18B20 (waterproof probe, IP67)	1-Wire (parasitic power capable)	Compensates TDS and pH readings; independent indicator of microbial growth rate; 5% of TankScore
Level sensor	JSN-SR04T ultrasonic (IP67)	Trigger/Echo GPIO	Water volume and refill detection; enables discrimination between source-water contamination and sludge resuspension; 10% of TankScore
Home pump-control unit	ESP8266 (NodeMCU) + 1.8″ ST7735 TFT + opto-relay module + AC-3 contactor + MCB	Wi-Fi or nRF24L01+; relay contact → contactor coil	Displays tank status indoors and switches the pump safely: opto-isolated relay drives a motor-rated contactor, with fail-safe OFF and one local override button
Power system	5–10 W solar panel + TP4056/BQ25895 charge controller + 18650/$LiFePO_4$ pack + buck converters (12 V/5 V/3.3 V)	—	Self-sustaining off-grid power; supports 2–3 weeks zero-sun autonomy; regulated rails for the battery bus, sensors and MCU
Enclosure	IP65 ABS/ polycarbonate box with cable glands, lid-mount brackets	—	Protects electronics from humidity and splash; mounts on tank lid; sensor cables exit through sealed glands
Software / firmware	ESP-IDF or Arduino framework; embedded HTTP server; LittleFS/ SPIFFS flash logging	Wi-Fi AP/STA	All edge logic and the local dashboard — constitutes the software-implemented portion of the IoT system

5.3 Sensor Subsystem

A plurality of sensors are positioned below the water line, suspended from the enclosure by waterproof cables through a sealed cable gland:

Sensor	Measurement Principle	Role in TankScore	Weight
Turbidity			40%

	Infrared light scattering (90° nephelometry)	Primary indicator of suspended particulate matter, biofilm fragments, and sludge resuspension	
Total Dissolved Solids (TDS) / Electrical Conductivity	Two-electrode conductivity measurement with temperature compensation	Indicator of dissolved chemical leaching from settled debris, scale formation, and source water changes	25%
pH	Glass electrode potentiometry	Indicator of biological activity (algae raise pH, anaerobic decay lowers pH), corrosion, and chemical contamination	20%
Temperature	Digital thermistor (DS18B20 or equivalent)	Used for temperature compensation of TDS and pH readings; also serves as an independent biological activity indicator (warmer water accelerates microbial growth)	5%
Water Level	Ultrasonic time-of-flight measurement	Determines water volume, detects refill events, identifies periods of no consumption, and enables detection of sludge resuspension during inlet flow	10%

5.4 How It Works — Operational Sequence

The invention operates as a continuous, autonomous sense–compute–alert–control cycle. The following sequence describes one complete operating cycle of a working embodiment:

> **STEP 1 — WAKE & POWER-UP.** The ESP32 wakes from deep sleep (default: every 3 hours) via RTC timer. It energises the 5 V sensor rail through a load switch, initialises the I²C bus (ADS1115), 1-Wire bus (DS18B20) and UART/GPIO for the level sensor. Deep-sleep current between cycles is <1 mA to conserve battery.

> **STEP 2 — ACQUIRE.** All five sensors are read in sequence: turbidity (analog), TDS (analog), pH (via ADS1115 16-bit ADC), temperature (1-Wire) and level (ultrasonic time-of-flight). Readings are repeated 8–16 times and median-filtered to reject outliers. A timestamp is attached to the sample.

> **STEP 3 — FUSE & SCORE (edge computation).** The processor normalises each parameter into a 0–100 sub-score, applies temperature compensation, then computes the weighted composite **TankScore** (turbidity 40% + TDS 25% + pH 20% + level 10% + temperature 5%). The sample is appended to a ring buffer in flash (last 112 samples ≈ 14 days).

> **STEP 4 — TREND & EVENT ANALYSIS.** Linear regression is run over the 14-day window for turbidity, TDS and pH. The system classifies the current event as (i) gradual degradation, (ii) source-water contamination (level rise + turbidity spike), or (iii) sludge resuspension (turbidity spike with no level change). This classification determines the score penalty and the recommended action.

> **STEP 5 — DECIDE.** The TankScore is mapped to a state: GOOD (80–100), WATCH (50–79), ACTION ADVISED (30–49) or CLEAN REQUIRED (0–29). The state drives a prioritised local alert and estimates days-to-clean from the trend slope; a "Tank Cleaned" confirmation is available on the dashboard after manual cleaning.

STEP 6 — PUBLISH. The tank unit publishes level, TankScore and state over the local link — Wi-Fi HTTP, or an optional nRF24L01+ radio link — to the indoor home unit every sample period.

STEP 7 — CONTROL (home unit). The home unit switches the pump through an opto-isolated relay and motor-rated contactor: ON when level falls below the low threshold, OFF above the high threshold, with manual override and a 10-minute auto-release. On processor reset or link loss beyond 60 s the relay is de-energised and the pump is forced OFF.

STEP 8 — REPORT & LOG. The event record (timestamp, TankScore, state, pump runs, cleaning confirmations) is appended to flash storage. The embedded web server updates the dashboard with current readings, the TankScore banner, trend charts and history. Battery/solar status is included.

STEP 9 — SLEEP. The sensor rail is de-energised, the ESP32 enters deep sleep, and the cycle repeats at the next timer interval.

5.5 IoT Subsystem (Highlighted)

The IoT subsystem is what converts a set of sensors and a pump into an autonomous, locally observable, edge-intelligent system. It comprises the following elements:

(a) Edge Intelligence — Computation On-Device, Not in the Cloud

All decision-making runs on the ESP32's dual-core Xtensa processor: sensor fusion, weighted scoring, 14-day linear-regression trend analysis, three-way event classification, alert decision logic and local publication of status. **Nothing is sent to a cloud server for analysis.** This means:

- Health decisions are made in real time on-device, immune to internet outages or server downtime.
- No user data leaves the premises — an architectural privacy guarantee.
- No recurring cloud cost, no account, no API key, no subscription.
- The device remains fully functional with zero internet connectivity.

(b) Local Wireless Communication & Dashboard

The ESP32 operates as a Wi-Fi access point (or joins an existing LAN in station mode) and hosts an embedded HTTP server. Any browser-equipped device on the same network — phone, tablet, laptop — can open a URL (resolved via mDNS, e.g. *tankguard.local*) and load the dashboard. The dashboard provides:

- Real-time gauges for turbidity, TDS, pH, temperature and level
- Colour-coded TankScore status banner (GOOD / WATCH / ACTION ADVISED / CLEAN REQUIRED)
- Trend charts at 24-hour, 7-day, 30-day and 90-day horizons
- "Tank Cleaned" confirmation button and cleaning history
- Battery and solar charging status; sensor calibration interface

Because the dashboard is served from the device itself, it works in buildings with no internet, in rural installations, and in situations where the owner does not want water-usage data held by a third-party cloud provider.

(c) Data Persistence at the Edge

All sensor samples and cleaning/pump event records are written to the ESP32's flash filesystem (LittleFS/SPIFFS) using wear-levelled append operations. The 14-day ring buffer supports trend regression; longer history supports the 90-day charts. Data survives power cycles and reboots. Optional export is available via a CSV download endpoint on the local server.

(d) Optional Extended IoT Connectivity (Claimed as Preferred Embodiment, Not Required)

In a preferred optional embodiment, the same ESP32 can additionally connect to the internet and push alerts through one or more channels — MQTT to a self-hosted broker, e-mail, SMS gateway, or a messaging-app bot — *without changing any core logic*, because the health decision is already made at the edge. The internet link is used only for notification and remote status viewing, never for control. This keeps the system's safety-critical function independent of connectivity.

(e) OTA Firmware Updates

An optional over-the-air firmware update mechanism allows the device's firmware to be improved after deployment (e.g. new TankScore weights, new sensor drivers) while the unit remains installed inside the tank — no physical removal required.

(f) Multi-Tank Fleet Management (Optional Embodiment)

Where multiple tanks exist on one site (apartment complex, school, hospital), each TankGuard unit advertises its own mDNS hostname and can be aggregated by a simple local dashboard that polls each device's HTTP endpoint — a lightweight local "fleet" view with no cloud platform involved.

5.6 TankScore Computation Engine

The TankScore is a composite numerical index (0–100) computed by the onboard processor. It integrates three classes of information:

(a) Instantaneous Parameter Sub-Scores

Each sensor parameter is converted to a 0–100 sub-score based on its distance from a safe reference band. For example, turbidity below 1 NTU scores 100; between 1–5 NTU scores linearly from 100 to 50; above 5 NTU scores 0.

(b) Trend Analysis

Linear regression is performed on the last 14 days (112 readings at 3-hour intervals) of turbidity, TDS, and pH data. The slope (rate of change) of each parameter contributes a trend bonus or penalty to the TankScore. A rising turbidity trend triggers a premature WATCH or CLEAN-NOW status even before the absolute threshold is breached, providing predictive capability.

(c) Event Classification

The system classifies water quality events into three categories based on correlated multi-parameter signatures:

- **Gradual degradation:** Turbidity slowly rising with stable pH → biofilm accumulation. TankScore penalized proportionally to slope.
- **Source water contamination:** Level rise (refill detected) followed by turbidity spike → contaminated incoming water. TankScore penalized based on peak turbidity but trend analysis is suppressed for settling period.
- **Sludge resuspension:** Turbidity spike without level change (inlet turbulence, temperature convection) → resuspension of existing bottom layer. TankScore penalized and a cleaning recommendation is issued.

TankScore Status Output

Status	TankScore Range	System Action
GOOD	80–100	No action required. Normal monitoring continues.
WATCH	50–79	Dashboard displays warning. Trend extrapolation shows estimated days-to-clean. User advised to plan cleaning.
ACTION ADVISED	30–49	System recommends cleaning soon and shows the evidence; user may confirm cleaning via the dashboard after it is done.
CLEAN REQUIRED	0–29	Persistent alert for manual cleaning; state remains until a "Tank Cleaned" confirmation or a sustained score recovery is recorded.

5.7 Mains-Isolated Pump-Control Subsystem

Cleaning remains a manual drain-and-scrub operation; the invention does not actuate anything on the tank itself. The pump-control subsystem is a separate indoor unit comprising:

1. **Opto-isolated relay:** A 5 V relay module with an optocoupler barrier, its contacts rated 10 A / 250 V AC, driven from an ESP8266 GPIO. The optocoupler keeps the low-voltage electronics galvanically separated from the switching circuit.
2. **Motor-rated contactor:** The relay contacts energise the coil of an AC-3 rated contactor sized for the pump's locked-rotor current. All pump mains wiring terminates at the contactor, never at the electronics board.
3. **Local display and control:** A 1.8″ ST7735 TFT shows tank level and TankScore; one physical button provides manual override (short press = toggle pump, long press = toggle AUTO/MANUAL), and manual mode auto-releases after 10 minutes.
4. **Fail-safe logic:** The relay is active only while the controller is healthy. On reset, crash or loss of the tank link for longer than 60 s, the relay de-energises and the pump stops. A miniature circuit breaker and the existing starter protection remain in the mains path.

Relay ON condition = link healthy AND (AUTO with level below LOW threshold OR manual override active)

This architecture satisfies the safety invariant that the embedded electronics never make or break mains current directly; the contactor, not the relay or the microcontroller, carries the pump load.

5.8 Power Management Subsystem

The unit is self-powered by a solar photovoltaic panel and rechargeable battery system:

- Solar panel: 5–10 W, 12 V nominal, mounted adjacent to or integral with the enclosure.
- Charge controller: MPPT or PWM, with overcharge and over-discharge protection.
- Battery: Lithium-ion (18650 or $LiFePO_4$) pack, capacity selected for 2–3 weeks of zero-sun autonomy.
- Regulated power rails: 12 V battery bus; 5 V rail for sensor excitation; 3.3 V rail for processor and digital circuits.
- Duty cycling: The processor enters deep-sleep mode between measurement cycles, waking at predetermined intervals (default: every 3 hours) to power the sensor rail, perform measurements, compute TankScore, publish status, and serve any pending user requests. Deep-sleep current is minimized (<1 mA) to maximize battery life.

5.9 User Interface Subsystem

The processor hosts a local web server accessible over the tank's WiFi network (no internet required). The interface provides:

- Real-time gauge display of all measured parameters
- TankScore status banner with color-coded indication
- Trend charts (24-hour, 7-day, 30-day, 90-day) for each parameter
- "Tank Cleaned" confirmation button (resets the cleaning interval counter and logs the event)
- Cleaning and pump-run history log
- Battery and solar charging status
- Calibration interface for sensor maintenance

6. INDEPENDENT CLAIM (DRAFT)

CLAIM 1 (Independent — System Claim):

A self-powered in-tank water quality monitoring and mains-isolated pump-control system for a water storage tank, the system comprising:

(a) an enclosure configured to be removably mounted on or within said water storage tank;

(b) a plurality of water quality sensors suspended from said enclosure and submerged in stored water within said tank, said sensors comprising at least a turbidity sensor, a total dissolved solids sensor, a pH sensor, a temperature sensor, and a water level sensor;

(c) a processor disposed within said enclosure, operatively connected to said sensors, configured to:

(i) periodically measure water quality parameters from said sensors;

(ii) compute a composite water quality score from said measured parameters, said score incorporating at least an instantaneous parameter component and a trend component derived from rate-of-change of said parameters over a predetermined historical window;

(iii) classify said composite score into one of a plurality of quality states; and

(iv) generate a prioritised cleaning alert when said composite score indicates a quality state requiring manual cleaning, and publish said score and state over said local-network interface;

(d) a separate pump-control enclosure, distinct from said tank-mounted enclosure, comprising an opto-isolated relay whose low-voltage input is driven by a controller output and whose contacts are configured to energise a coil of a motor-rated contactor, said contactor being arranged to connect an electric pump to a mains supply, whereby the relay provides galvanic isolation between the controller and the mains switching circuit and the contactor carries the pump load;

(e) a local override input on said pump-control enclosure and a fail-safe circuit configured to de-energise said relay upon controller reset or loss of communication with said tank-mounted enclosure for a predetermined timeout, thereby forcing the pump OFF; and

(f) a solar photovoltaic panel and rechargeable battery disposed within or adjacent to said tank-mounted enclosure, said solar panel and battery providing electrical power to said sensors and said processor; and

(g) an IoT communication subsystem comprising a wireless local-network interface hosted by said processor, said interface comprising an embedded HTTP server and a multicast-DNS responder, configured to serve an interactive user dashboard to a client device on the same local network without requiring internet connectivity, cloud infrastructure or a user account,

wherein said processor is further configured to execute all of said measurement, scoring, classification and alerting logic at the network edge on said processor without transmitting sensor data to any remote server, and to store a local event log, including cleaning confirmations and pump-run events, for presentation on said user dashboard.

CLAIM 7 (Dependent — Edge/IoT Architecture):

The system of Claim 1, wherein said composite water quality score, said trend analysis, said event classification and said alert decision are all computed locally on said processor within said enclosure, and wherein the system operates its full sense–compute–alert–control loop without any internet connection, cloud service, subscription or user account; and wherein said local wireless network interface is the sole means by which sensor data is presented to a user.

CLAIM 8 (Dependent — Duty-Cycled Edge Sensing):

The system of Claim 1, wherein said processor operates in a low-power sleep state between measurement cycles and wakes at predetermined intervals to energise a switched sensor power rail, acquire sensor readings, compute the composite score, publish status and serve pending dashboard requests, thereby enabling solar-battery operation with multi-week autonomy while maintaining continuous logical monitoring of the stored water.

7. CLAIMS FOR NOVELTY ASSESSMENT

#	Claimed Feature	Novelty Rating	Justification
1	Trend-predictive TankScore with event classification	**HIGH**	Prior art monitoring systems use simple threshold alerts. The composite score with 14-day slope analysis, acceleration detection and three-way event classification (degradation vs refill contamination vs resuspension) is a novel computational method.
2	Sludge resuspension detection (turbidity spike without level change)	**MEDIUM-HIGH**	This specific event classification — distinguishing resuspension from source contamination by correlating turbidity with level data — is not found in surveyed prior art.

3	Edge-only IoT architecture (no cloud) for tank monitoring	**MEDIUM-HIGH**	All surveyed IoT water monitoring systems rely on cloud platforms for analysis and dashboards. Here the entire control loop — sensing, scoring, trend regression, event classification and alerting — executes on the device, with a self-hosted HTTP/mDNS dashboard. Deliberate architectural distinction with practical benefits (privacy, cost, offline operation, outage immunity).
4	Duty-cycled edge sensing with switched sensor power rail (solar autonomy)	**MEDIUM-HIGH**	Combining deep-sleep MCU duty cycling with switched sensor excitation rails to achieve multi-week solar autonomy while maintaining a continuous logical monitoring loop is a specific power-architecture contribution not disclosed in surveyed prior art.
5	Two-unit architecture with mains-isolated pump control and fail-safe link timeout	**MEDIUM**	Pump controllers and tank level monitors exist separately, but the combination of edge-computed tank health, a separate opto-relay/contactor pump stage and a forced-OFF fail-safe on link loss is not disclosed in the surveyed references.

8. PRIOR ART SEARCH SUMMARY

Reference	Title / Assignee	Relevance	Distinction from Present Invention
CN223988161U	Automatic sludge discharge control system (Chengdu, China)	Industrial sludge discharge	Uses camera-based image analysis, not water quality sensors. Designed for municipal sedimentation tanks (industrial scale), not domestic storage tanks. No composite water-quality scoring or trend prediction.
IN202311028143	Smart Water Tank Management System (Uttaranchal University, India)	Tank quality monitoring with sensors	Sensor-only system. Alerts the user via a mobile app with raw readings or simple thresholds; no composite trend-aware score, no pump control, no local-network dashboard.
IN202441068267	Automatic Cleaning of Overhead Water Tanks (SR University, India)	Automated tank cleaning	Uses a mechanical wiper to scrub walls + chlorine doser + full drain. Requires manual initiation and does not use in-tank water quality sensors to decide when cleaning is due; no predictive scoring.
IN (Marwadi University)	Water Purity Monitoring and Automated Purification Device	Sensing + treatment	Tests water in a separate chamber (not in-tank), then dispenses chemical treatments. Complex multi-chamber design; no continuous in-tank trend monitoring and no pump control.
ILONNATI Innovations	Smart Intelligent Water Monitoring	IoT sensors + motorized valves	Designed for piped water supply networks, not storage tanks. Monitors supply quality at

			multiple network points. No in-tank sensing and no pump control.
WO2024173389A1	Water Quality Monitoring System (AS America Inc, USA)	Multi-sensor monitoring + phone app	Pure monitoring system with no pump control or local dashboard. Monitors water in pipes, not in storage tanks.
US (Delta Faucet)	Smart Water Sensing and Management System	Sensors + valve actuation	Monitors water at point-of-use (faucets/ showers) using ORP and conductivity. Actuates valves to purge pipe water; not a tank monitor and not a pump controller. Tiered purge logic for plumbing systems, not stored-water tanks.

9. PATENTABILITY ASSESSMENT

Novelty (Section 32 of Indian Patents Act, 1970): The claimed subject matter is not anticipated by any single prior art reference found in the preliminary search. The combination of in-tank multi-parameter sensing, trend-predictive composite scoring with event classification, a cloud-free local dashboard, and a separate mains-isolated pump controller with fail-safe link timeout is not disclosed in any surveyed reference.

Inventive Step / Non-Obviousness: While each individual component (ESP8266/ESP32, sensors, relay, contactor, solar panel, embedded web server) is known in isolation, their specific combination is not obvious to a person skilled in the art of water tank maintenance or IoT sensor systems. In particular: (a) the computational method distinguishes settled-sediment resuspension from source-water contamination at the network edge by correlating turbidity with level changes and trend slope, rather than using a single threshold; (b) the entire health-assessment loop runs on-device with a self-hosted dashboard, eliminating the cloud dependency universal in prior-art IoT water monitors; and (c) pump control is deliberately split into a separate unit with an opto-isolated relay, a motor-rated contactor and a forced-OFF fail-safe on link loss, satisfying a safety constraint that prior-art monitor-plus-app systems do not address.

Industrial Applicability: The invention is directly applicable to millions of domestic and institutional water storage tanks in India and similar regions. The system uses commercially available components and can be manufactured at low cost. The self-powered, cloud-free design makes it suitable for deployment in areas with unreliable electricity and internet connectivity.

10. RECOMMENDED CLAIM STRATEGY

1. **Independent System Claim (Claim 1):** Broadest claim covering the complete integrated system. Drafted above.
2. **Dependent Claim — Trend-Predictive Alerting:** The system of Claim 1, wherein the composite score incorporates a trend component derived from linear regression analysis of historical turbidity and TDS readings over a period of 7 to 30 days, and wherein the cleaning alert is generated when the trend component indicates a

rate of degradation that would cause the instantaneous parameters to exceed safe thresholds within a predetermined number of days, even if the instantaneous parameters have not yet exceeded said thresholds.

3. **Dependent Claim — Event Classification:** The system of Claim 1, wherein the processor classifies water quality events into a plurality of event types based on correlated multi-parameter signatures, said event types comprising at least: gradual degradation, source-water contamination, and sludge resuspension, and wherein each event type produces a different TankScore contribution and a different recommended remediation action.
4. **Dependent Claim — Mains Isolation and Fail-Safe:** The system of Claim 1, wherein the pump-control enclosure comprises an optocoupler barrier between the controller and the relay coil, and a watchdog configured to de-energise the relay and stop the pump when communication with the tank unit is lost for more than 60 seconds or when the controller resets.
5. **Dependent Claim — Edge/IoT Architecture:** The system of Claim 1, wherein all scoring, trend analysis, event classification and alerting logic execute locally on the on-device processor, with the full control loop operating without internet, cloud, subscription or account, and the local wireless interface being the sole user-facing data path. (Claim 7 above.)
6. **Dependent Claim — Duty-Cycled Edge Sensing:** The system of Claim 1, wherein the processor sleeps between measurement cycles and wakes on timer to energise a switched sensor rail, acquire, compute, decide and serve requests, enabling multi-week solar autonomy. (Claim 8 above.)
7. **Method Claim:** A method of monitoring water quality in a storage tank and controlling a pump using an IoT edge device, comprising: waking from low-power sleep; energising a sensor power rail; measuring turbidity, TDS, pH, temperature and water level in stored water; computing a composite water quality score with trend analysis at the edge; classifying the score into quality states; raising a prioritised cleaning alert when the score indicates cleaning is due; publishing the score, state and level to a separate indoor controller; switching the pump through an opto-isolated relay and a motor-rated contactor in dependence on the published level and a manual override input; forcing the pump OFF when the link is lost for a predetermined timeout; logging the event locally on the device; serving the results on a local-network dashboard; and returning to sleep.

— End of Patentability Disclosure —

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